LEARNABILITY OF FROBENIUS TRACES: COMPLEX MULTIPLICATION VERSUS GENERIC ELLIPTIC CURVES

Abstract. Recent work applies machine learning to the Frobenius traces a_p of elliptic curves, predicting one trace from the others and observing that the learned embeddings organise themselves by residue modulo small integers. We isolate the arithmetic source of this learnability with a controlled experiment. Comparing a family of CM curves (CM by $\mathbb{Z}[\omega]$) against a family of generic non-CM curves, a small feed-forward model predicts $a_p \bmod 2$ at split primes with a mean accuracy lift of $+0.12$ over the majority-class baseline for the CM family, and -0.12 (no signal) for the generic family. When each CM field is matched to its own reciprocity character—$\chi_3 \pmod 3$ for $\mathbb{Q}(\omega)$, $\chi_4 \pmod 4$ for $\mathbb{Q}(i)$—both CM families show lifts near $+0.19$ while generic families stay negative. We trace the gap to an exact class-field fact: for the conductor-27 CM curve $E : y^2 + y = x^3$, the trace satisfies $|a_p| = |L|$ where $4p = L^2 + 27M^2$, verified for all split primes below 500. The learnability of CM curves is therefore not mysterious; it is the cubic-residue splitting law χ_3 made visible to a statistical model.

1. Background and question

For an elliptic curve $E/\mathbb{Q}$ and a prime p of good reduction, write $a_p = p + 1 - \#E(\mathbb{F}_p)$. Recent work trains neural models to predict one a_p from a vector of others $(a_p[q])_{q \neq p}$ and finds, first, that $a_p \bmod 2$ is predictable above chance, and second, that the learned token embeddings separate by residue modulo 2, 3, 4 and 6 [LEF]. The mod-2 predictability has a clean explanation: a_p is even iff the mod-2 Galois representation is reducible, iff the 2-division polynomial has a root modulo p — a splitting (reciprocity) condition. We ask a sharper question: *is the reciprocity structure of a CM curve more learnable than that of a generic curve, and if so, which arithmetic fact is being learned?*

2. The experiment

We build three families of 60 curves each, evaluated at the 64 primes $5 \leq p < 320$: $\mathbf{CM}_\omega$: $y^2 = x^3 + D$ ($j = 0$, CM by $\mathbb{Z}[\omega]$); $\mathbf{CM}_i$: $y^2 = x^3 + Dx$ ($j = 1728$, CM by $\mathbb{Z}[i]$); and **generic**: $y^2 = x^3 + ax + b$ with a, b random, $ab \neq 0$ (non-CM). For a target prime p_*, the input is $(a_p[q] \bmod 2)_{q \neq p_*}$ and the label is $a_p[p_*] \bmod 2$; a two-layer feed-forward network (width 64, ReLU) is trained on 75% of each family and tested on the rest, averaged over 8 splits. We report test accuracy minus the majority-class baseline.

Supersingular columns are removed. The CM_ω family has $a_p = 0$ for every $p \equiv 2 \pmod 3$, and CM_i has $a_p = 0$ for every $p \equiv 3 \pmod 4$. At these inert primes $a_p \bmod 2 \equiv 0$ is constant, so prediction there is trivial and would inflate the score. We therefore evaluate CM_ω only at split primes $p \equiv 1 \pmod 3$, where $a_p \neq 0$ and $a_p \bmod 2$ is genuinely non-constant.

Remark 1 (Why CM_i is excluded from the headline)**.** The family $\mathrm{CM}_i : y^2 = x^3 + Dx$ has the rational 2-torsion point $(0, 0)$ for every D, forcing a_p even at all good primes. Thus $a_p \bmod 2 \equiv 0$ across the family and the task is vacuous (baseline 1.0, lift 0). This is a feature of the chosen model, not of $\mathbb{Z}[i]$: the correct character for $\mathbb{Z}[i]$ is χ_4, and the natural target is

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Family	baseline	test acc.	lift	95% CI
CM_ω (CM by $\mathbb{Z}[\omega]$)	0.715	0.835	$+0.121$	± 0.031
generic (non-CM)	0.639	0.519	-0.120	± 0.032

TABLE 1. a_p mod 2 at split primes. The CM family is learnable above baseline; the generic family shows no signal (a negative lift means the model underperforms majority-class guessing on held-out curves). Two-layer network (width 64), 60 curves per family, 64 primes; each lift aggregated over 8 target primes and 8 random splits ($N = 64$ for CM_ω, 88 for generic), 95% CI from the standard error.

a_p mod 4 at $p \equiv 1 \pmod 4$ (Fermat's two-square law $p = a^2 + b^2$). We record this because it is exactly the kind of trivial-column effect that could be mistaken for a null result.

3. The arithmetic behind the gap

The learnability gap is a theorem, not a black-box phenomenon.

Proposition 2. *For $E : y^2 + y = x^3$ (conductor 27) and a prime $p \equiv 1 \pmod 3$, write $4p = L^2 + 27M^2$ (unique primitive representation up to signs). Then $|a_p| = |L|$. Verified for all 45 split primes below 500.*

This is the Gauss–Deuring description of CM Frobenius: $a_p = \pi + \bar{\pi}$ for a primary generator $\pi \in \mathbb{Z}[\omega]$ of the prime above p, with $\pi + \bar{\pi} = 2a - b$ in the basis $\pi = a + b\omega$. So a_p for a CM curve is a function of how p is represented by a quadratic form — a class-field datum depending only on p and the field, not on fine features of the curve. A generic curve has no such law; its a_p requires the individual curve. This is the precise reason a model finds structure in the CM family and not the generic one: the bit a_p mod 2 is one bit of L, and L is pinned by χ_3 and $4p = L^2 + 27M^2$.

Observation 3. The experiment inverts the classical computation. Where one usually derives a_p from the reciprocity law of $\mathbb{Q}(\omega)$, here a model recovers the fact that CM curves carry a learnable reciprocity structure generic curves lack. Both directions concern the same object, the splitting law χ_3, and agree.

4. Matched characters: χ_3 versus χ_4

Remark 1 predicts that the vacuous mod-2 result for $\mathbb{Q}(i)$ is an accident of forced 2-torsion, and that the right target for a CM field is the character that field carries. We test this directly, matching each family to its own reciprocity character and evaluating at that character's split primes.

The two exceptional imaginary-quadratic fields—those with extra units, $w = 6$ and $w = 4$—give the same learnable signal once the character is matched, and the generic controls stay negative throughout. This is the thesis in its sharpest form: the model learns the reciprocity law of the CM field, and only when asked about the character that field actually carries.

Conjecture 4. Let E have CM by an order in an imaginary quadratic field K with quadratic character χ_K. Then a_p mod m is learnable above the majority-class baseline (from $(a_p[q] \bmod m)_{q \neq p}$ across a CM family) precisely when m is divisible by the conductor of χ_K, with a lift bounded away from zero as the family and prime range grow. Generic (non-CM) families exhibit no comparable asymptotic signal.

Field	units	character	target	lift	95% CI
$\mathbb{Q}(\omega)$	6	χ_3	a_p mod 3	$+0.186$	± 0.029
$\mathbb{Q}(i)$	4	χ_4	a_p mod 4	$+0.189$	± 0.037
generic	—	—	a_p mod 3	-0.139	± 0.028
generic	—	—	a_p mod 4	-0.135	± 0.028

Table 2. Each CM field matched to its own character. Both CM families are learnable at nearly identical strength ($+0.19$); both generic controls are negative. Notably $\mathbb{Q}(i)$, vacuous on the mod-2 target (Remark 1), becomes strongly learnable on its correct target a_p mod 4 (Fermat: $p = a^2 + b^2$ at $p \equiv 1 \pmod 4$). Same protocol as Table 1.

5. Scope

This is a demonstration on small families (60 curves, 64 primes), not a large-scale study; a publication-grade estimate would use 10^4 curves. The CM_ω family is specifically $j = 0$. The matched-character comparison of §4 realises the $\mathbb{Q}(\omega)/\mathbb{Q}(i)$ pair on small families; scaling to 10^4 curves and confirming the conjecture's asymptotic claim is the natural sequel. No claim is made about additive questions: the learnable content is the multiplicative splitting law; the distribution of primes lies outside it.

References

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